

ELEC5305 Project Proposal

Speech and Music Presence Detection in Overlapping Audio

Using Audio Features and YAMNet

The University of Sydney

ELEC5305 – Acoustics, Speech and Signal Processing

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GitHub Repository:

<https://github.com/49sky/elec5305-project-530582937>

1. Project Overview

This project will investigate the detection of speech and music when both sources occur simultaneously in an audio recording. Instead of treating speech and music as two mutually exclusive classes, the system will independently determine whether speech is present and whether music is present.

Research question: *How does the relative level of speech and music affect the ability to detect both sources in an overlapping mixture, and how do interpretable audio features compare with a pretrained audio classifier as one source becomes progressively weaker?*

The main experimental variable will be the speech-to-music ratio (SMR), defined as:

$$\text{SMR} = 10 \log_{10}(P_{\text{speech}} / P_{\text{music}})$$

where the speech and music power terms represent the signal powers of the two sources. Controlled mixtures will be generated at several SMRs so that detection performance can be measured as either speech or music becomes weaker.

Two approaches will be compared. The first will use short-time signal-processing features with two simple linear support vector machines (SVMs). The second will use pretrained YAMNet as a modern reference system. The project will remain primarily implemented in MATLAB.

2. Background and Motivation

Automatic speech–music discrimination has been studied for many years. Earlier systems demonstrated that relatively simple audio characteristics such as energy, zero-crossing rate and spectral features can provide effective separation between isolated speech and music (Scheirer & Slaney, 1997). Similar low-level temporal and spectral features have also been widely used in music information retrieval and audio classification (Tzanetakis & Cook, 2002).

However, isolated speech-versus-music classification does not represent many real audio environments. Speech and music often occur at the same time in podcasts, television, advertisements, films and online videos. Recent work has therefore examined the more difficult problem of detecting speech and music under overlap. Bhattacharjee, Prasanna and Guha (2023) reported that clean speech/music discrimination is substantially easier than detecting the same sources when they overlap, motivating a controlled investigation of source level in this project.

The project will use the MUSAN corpus as the main source of speech and music recordings (Snyder, Chen, & Povey, 2015). A modern reference will also be provided by YAMNet, which is trained on AudioSet and produces scores for 521 audio-event classes, including speech and music-related categories (Gemmeke et al., 2017; Hershey et al., 2017). MATLAB provides direct access to pretrained YAMNet, so the project can compare a simple interpretable signal-processing system with a modern pretrained model without training a neural network from scratch.

3. Proposed Methodology

The project will be implemented mainly in MATLAB using speech and music recordings from MUSAN.

The original source recordings will first be divided into separate training, validation and test sets before segmentation or mixture generation. All segments derived from one original recording will remain in the same partition. This prevents source leakage between training and testing and provides a more reliable evaluation.

Speech and music segments will then be combined according to:

$$x[n] = s[n] + a m[n]$$

where $s[n]$ is the speech signal, $m[n]$ is the music signal, and a controls the relative music level. Mixtures will be generated at approximately -20 , -10 , -5 , 0 , $+5$, $+10$ and $+20$ dB SMR. Pure speech and pure music examples will also be retained as reference conditions.

The task will use two independent binary outputs:

Audio condition	Speech present	Music present
Speech only	1	0
Music only	0	1
Speech + music	1	1

For the handcrafted system, each audio segment will be divided into short overlapping frames. Features will include RMS energy, zero-crossing rate, spectral centroid, spectral flux, spectral rolloff, spectral bandwidth, spectral flatness and Mel-frequency cepstral coefficients (MFCCs). STFT spectrograms will also be used to examine representative mixtures.

Frame-level features will be summarised over each complete analysis segment using statistics such as mean and standard deviation. The aggregated feature vector will then be supplied to two linear SVM classifiers: one for speech presence and one for music presence.

The second system will use pretrained YAMNet in MATLAB. YAMNet will not be retrained or fine-tuned. Instead, speech- and music-related output scores will be extracted from the same test mixtures and compared against the handcrafted system.

An initial proof-of-concept will first use approximately 10 speech recordings, 10 music recordings and three SMRs of -10 , 0 and $+10$ dB. If the mixture generation, spectrograms and YAMNet scores behave sensibly, the experiment will then be expanded to the full set of SMRs and a larger source collection.

4. Expected Outcomes

The main outcome will be a reproducible MATLAB system that creates controlled speech–music mixtures and measures how reliably each source can still be detected as its relative level changes.

Performance will be evaluated separately for speech and music using precision, recall and F1-score. The most important results will be plots of speech recall and music recall against SMR for both the handcrafted SVM system and YAMNet. A further metric will be weaker-source recall, which measures whether the lower-power source remains detectable as it becomes progressively weaker.

The project is expected to show how overlap and relative acoustic level affect source detectability, although the exact behaviour will be determined experimentally rather than assumed in advance. Representative successes and failures will also be inspected using spectrograms, audio features and YAMNet scores to identify which acoustic characteristics make the weaker source easier or more difficult to detect.

5. Timeline

Week	Task
Weeks 1–2	Activity described in previous proposal
Weeks 3–5	-

Weeks 6–7	-
Weeks 8–9	Implement audio feature extraction and two linear SVM presence detectors.
Week 10	Implement pretrained YAMNet inference and extract speech/music scores.
Week 11–12	Run SMR experiments, calculate metrics, produce detection curves and analyse failure cases.
Week 13	Final report, GitHub documentation and reproducibility testing.

6. References

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